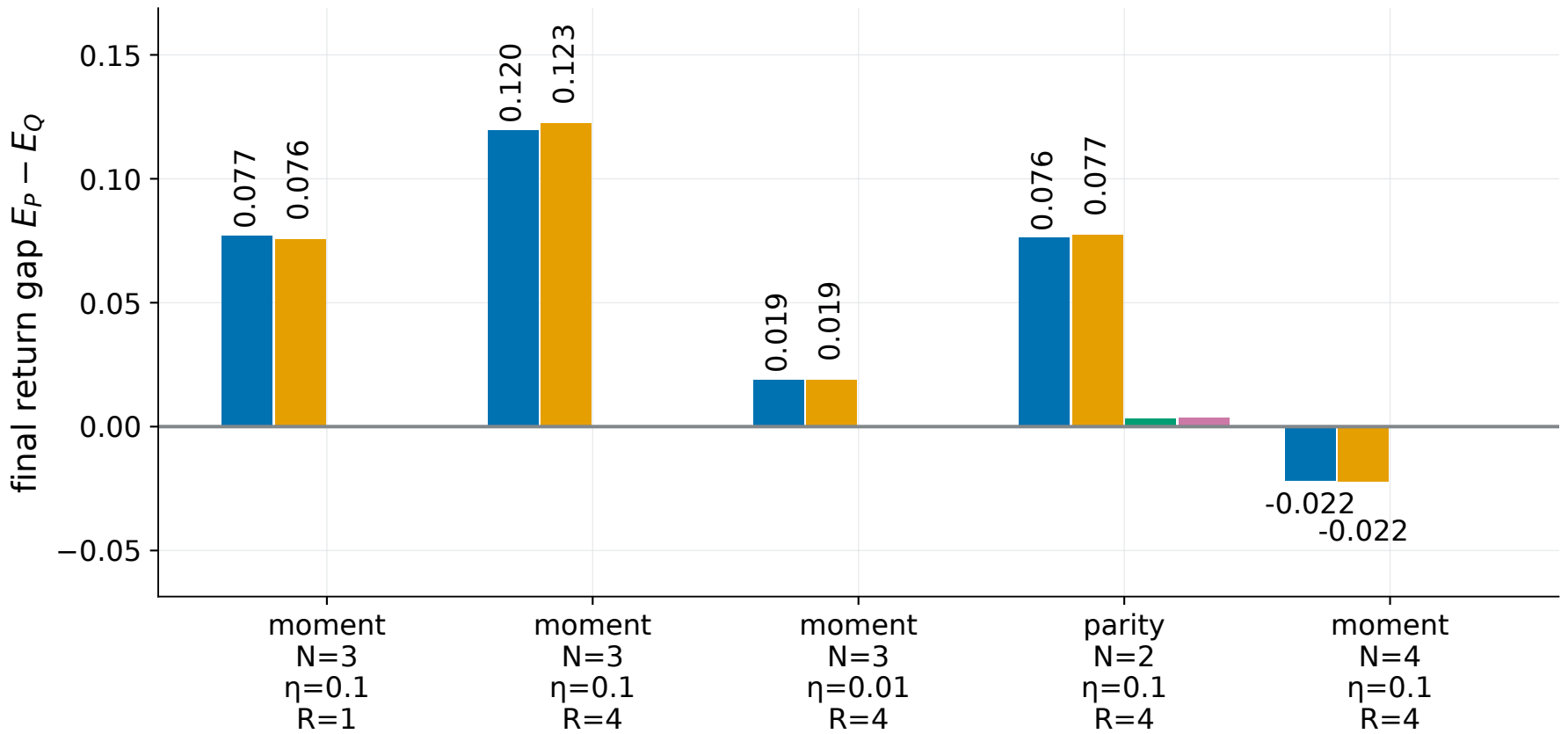
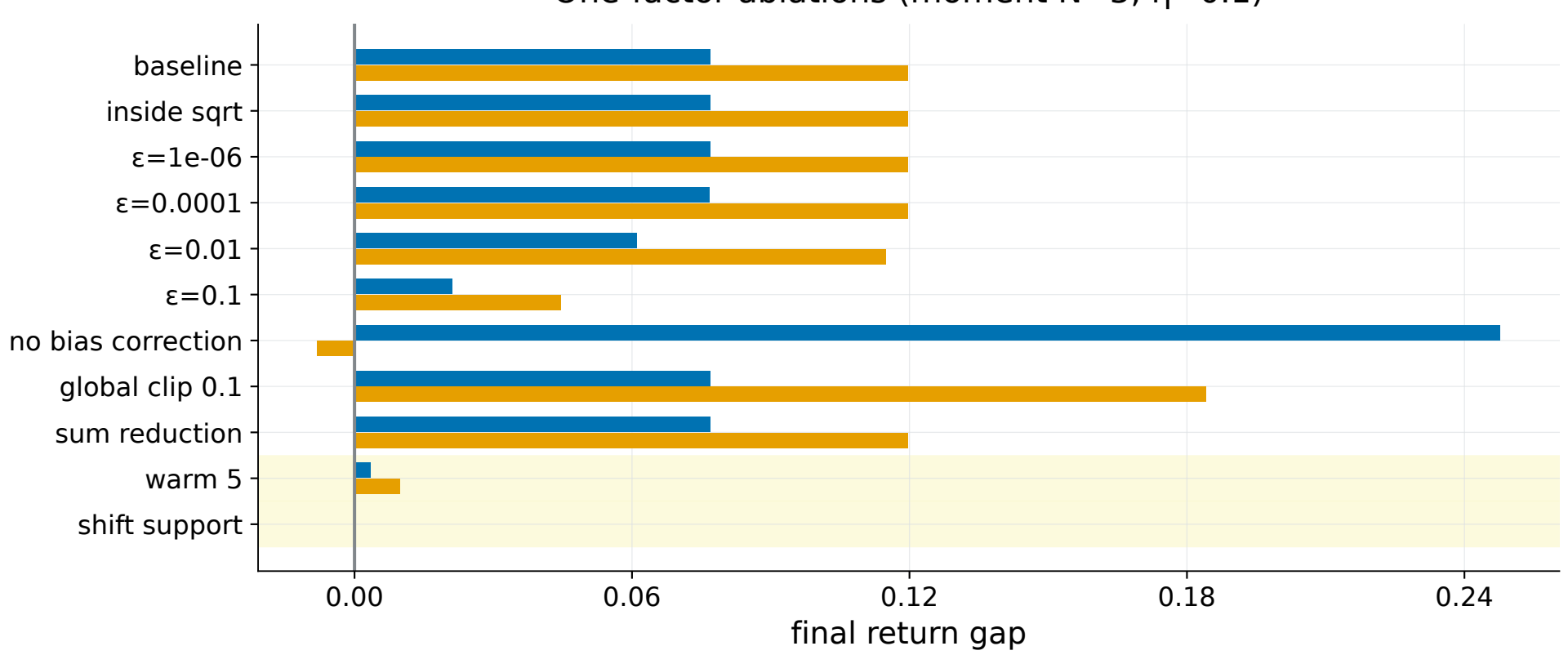
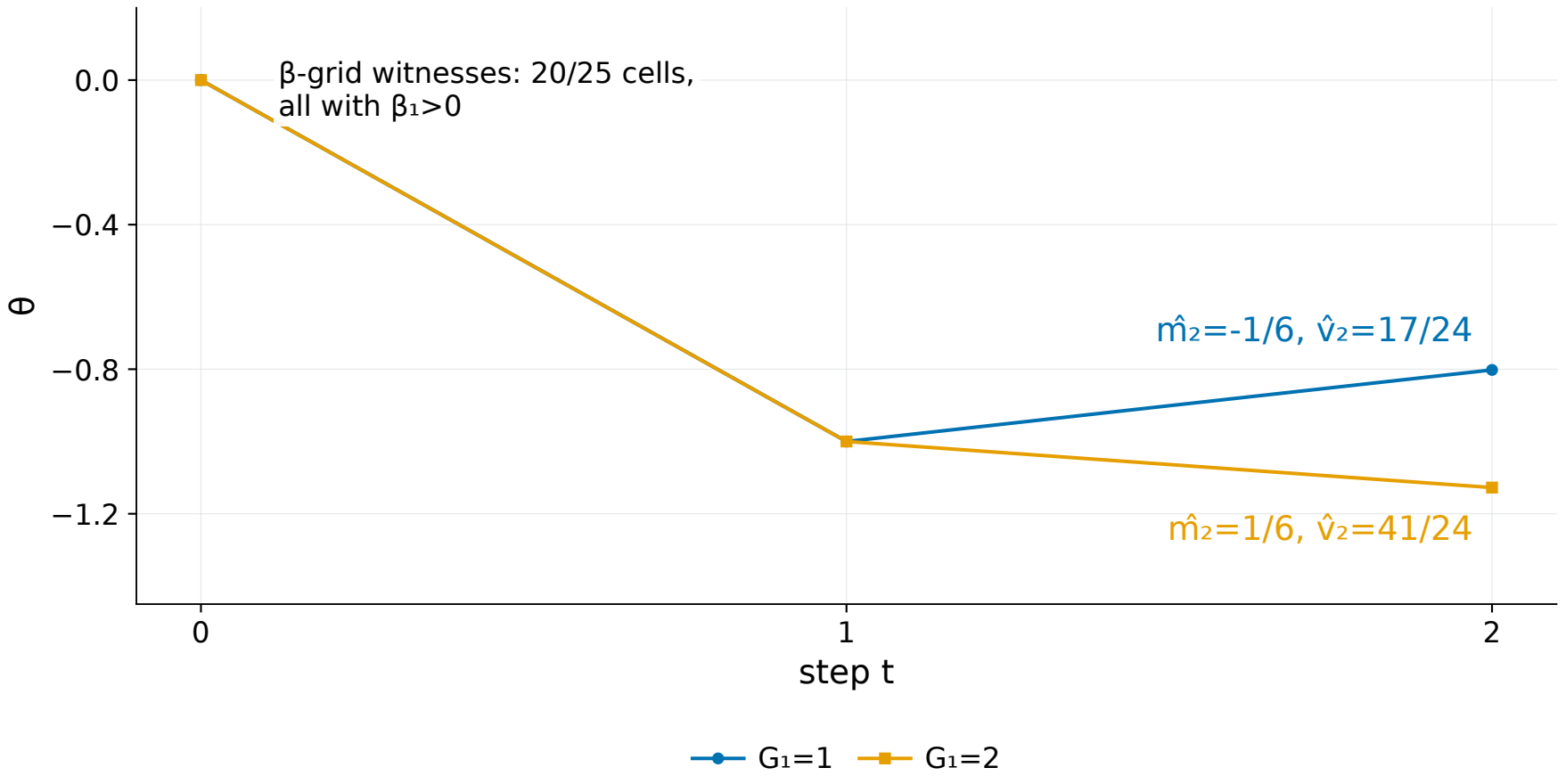
Same moments, different Adam response



One-factor ablations (moment N=3, η=0.1)



Same signs, different memory



Controlled small network and exact optimizer-memory witness; not a real-model result.